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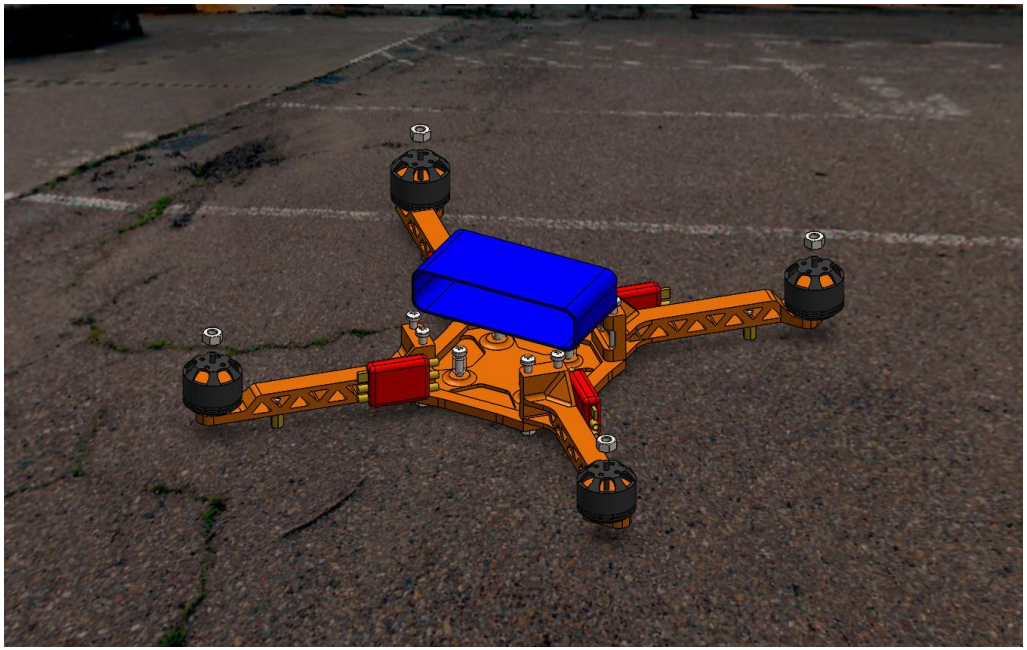
5-inch FPV-style quadcopter | CAD, FEA, engineering drawings, PID simulation, circuit schematic

- **CAD:** Complete frame assembly modelled in SolidWorks 2025 (reference geometry from GrabCAD) with a 13-item BOM and drawings.
- **Mass:** Estimated total about 390 g (about 426 g all-up with props and wiring).
- **FEA:** Static study on the arm. Rib-thickness sensitivity: 3.0 mm ribs pass (FOS 2.04), 2.0 mm ribs fail (FOS 1.08). Baseline retained.
- **Control:** Single-axis roll PID simulation in Python (P vs PD vs PID, with a wind gust).
- **Electronics:** Wokwi circuit: Arduino Uno + MPU6050 IMU + 4 ESC placeholders.

Key result	Value
Arm min. factor of safety (3.0 mm ribs)	2.04 (criterion: 2.0)
Arm max von Mises stress	14.72 MPa
PID gust final error (Kp 0.2, Ki 0.8, Kd 0.03)	0.03 deg
Estimated total mass	about 390 g

Software-only project: designed, modelled and simulated. Hardware not built.

Tools: SolidWorks Premium 2025 (Part, Assembly, Drawing, Simulation), Python (Google Colab), Wokwi.

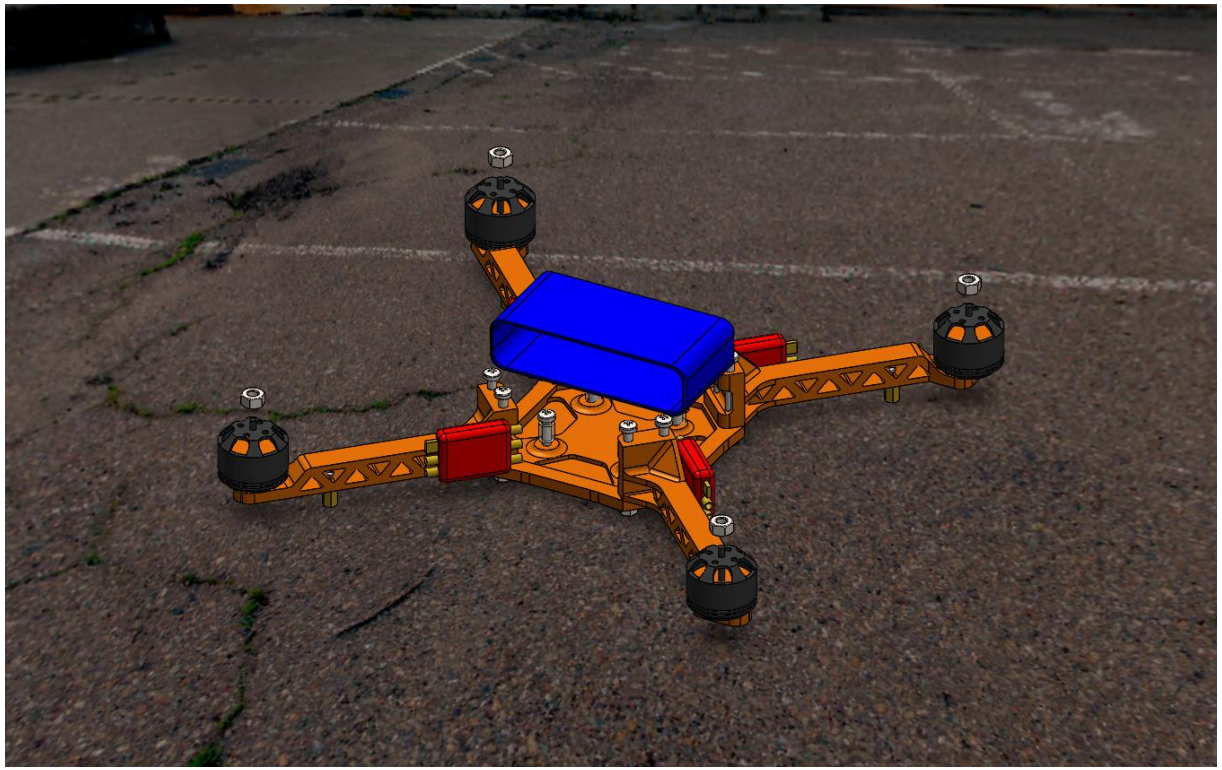


Isometric render

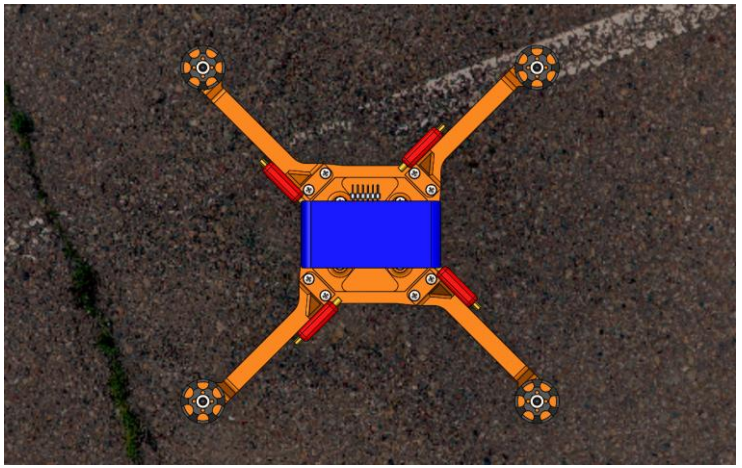
© 2026 Geethanjali Raguram | Quadcopter Design & Simulation | Reference geometry from GrabCAD | Hardware not built

CAD Model and Renders

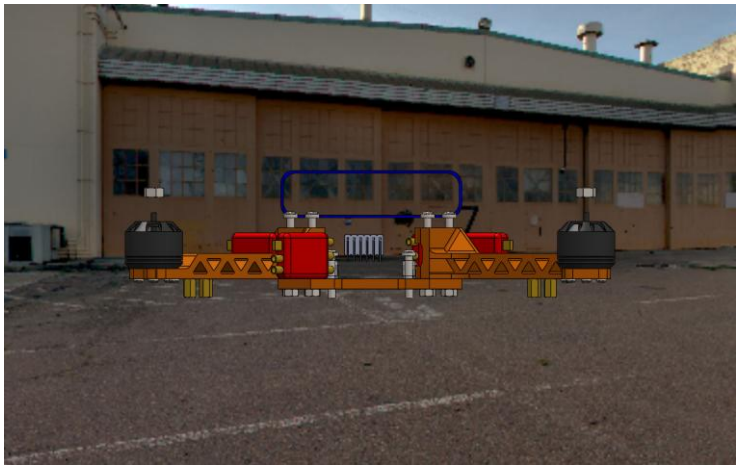
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Isometric view



Top view

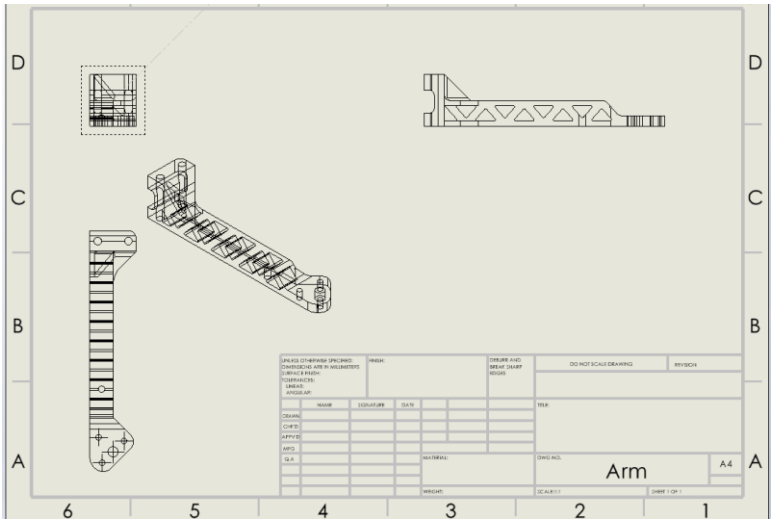


Front view

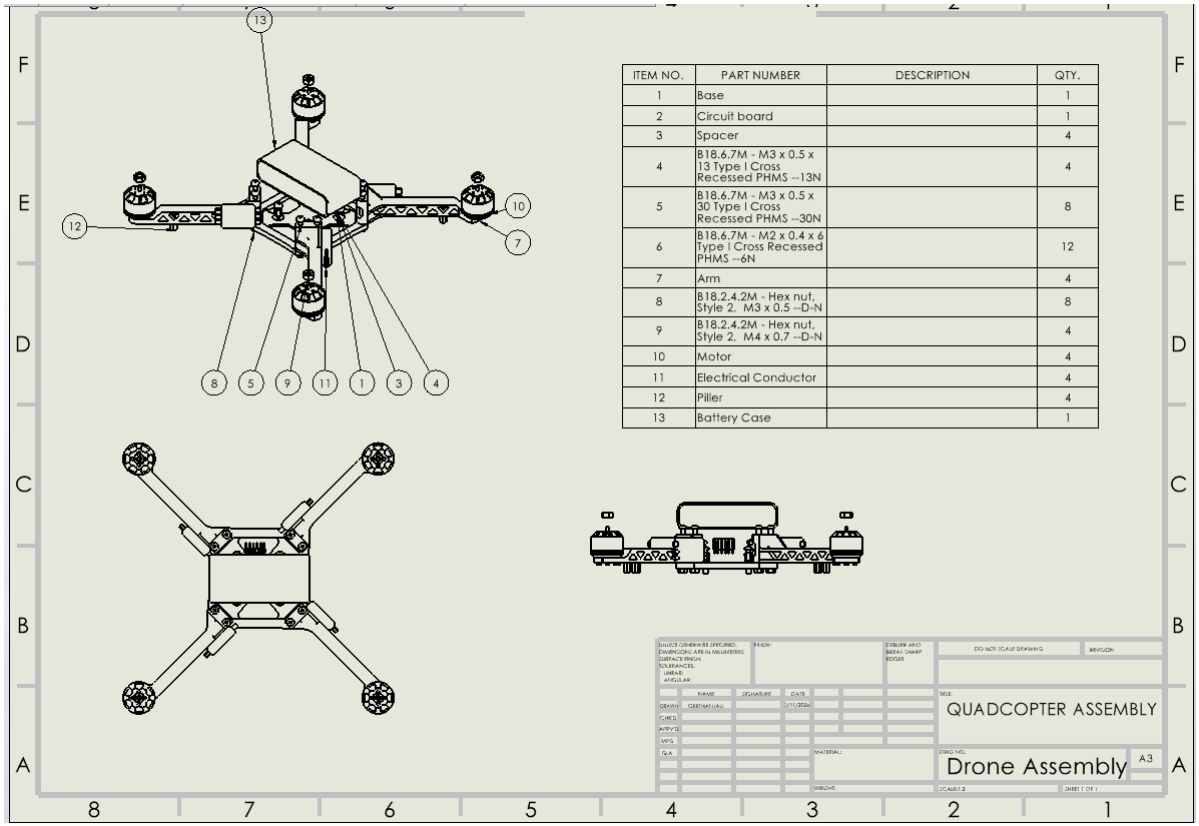
Parts: base plate, 4 arms, 4 motors, 4 electrical conductors, 4 pillars, spacer, circuit board, battery case, fasteners. Reference geometry from GrabCAD.

Engineering Drawings and BOM

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Arm part drawing (A4)



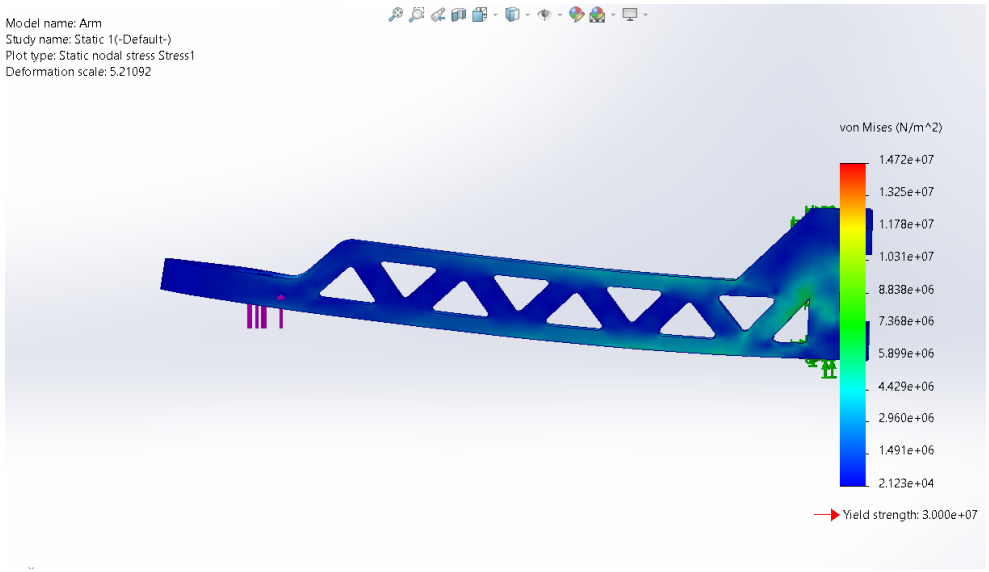
Assembly drawing (A3): isometric, top and front views, BOM, balloons

Note: the top plate, spinner and propeller have unresolved references in the assembly, so they are not in the BOM or drawing views.

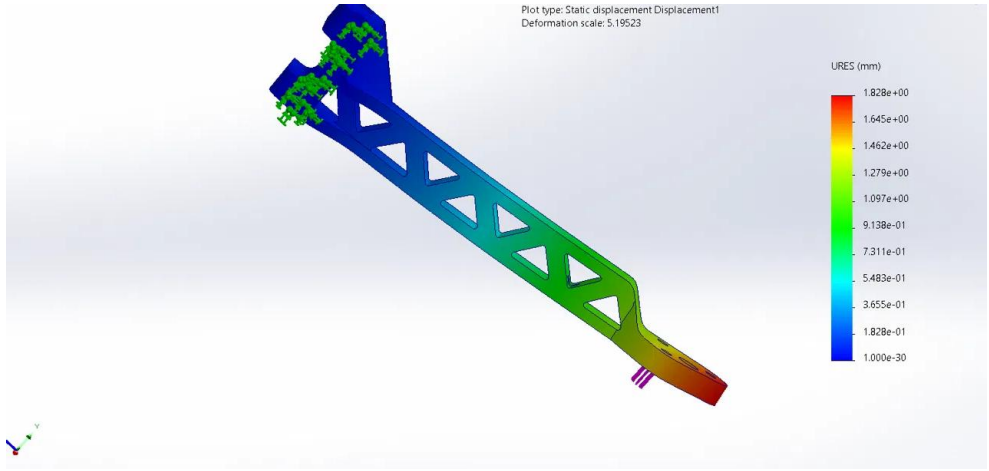
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Arm FEA: Baseline Design (3.0 mm ribs)

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von Mises stress. Max 14.72 MPa (assumed yield 30 MPa)



Resultant displacement. Max 1.828 mm (deformation scale about 5.2x, exaggerated)

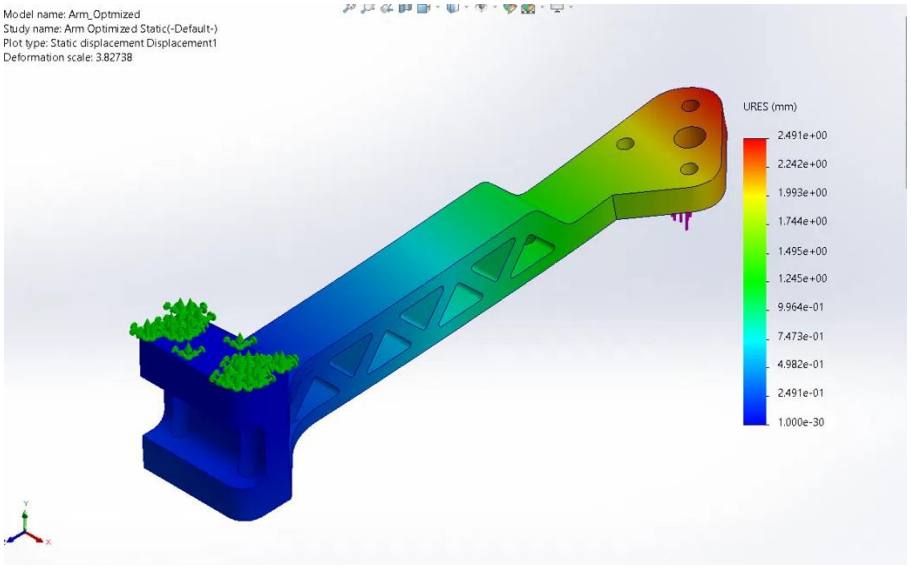
- **Setup:** ABS (E 2.41 GPa, Poisson 0.39, density 1070 kg/m3, yield assumed 30 MPa). Lug end face fixed. 10 N per motor at the motor end. About 80,941 elements, Direct Sparse solver.
- **Result:** Min FOS 2.04, which meets the criterion FOS >= 2. The peak stress is at the fixture edge, so it is partly artificial.

Rib-Thickness Sensitivity Study

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Metric	3.0 mm (baseline)	2.0 mm (trial)	Change
Mass	8.074 g	7.445 g	-7.8 %
Volume	7546.17 mm3	6957.80 mm3	-7.8 %
Max von Mises stress	14.72 MPa	27.8 MPa	+89 %
Max displacement	1.828 mm	2.491 mm	+36 %
Min factor of safety	2.04	1.08	-47 %
Decision	Retained (pass)	Rejected (fail)	

- Thinning the ribs saved only 0.63 g per arm but cut the safety margin below the FOS >= 2 criterion.
- The 3.0 mm design is kept. An intermediate 2.5 mm trial was not run.
- Single static load case and assumed yield strength, so results are indicative only.



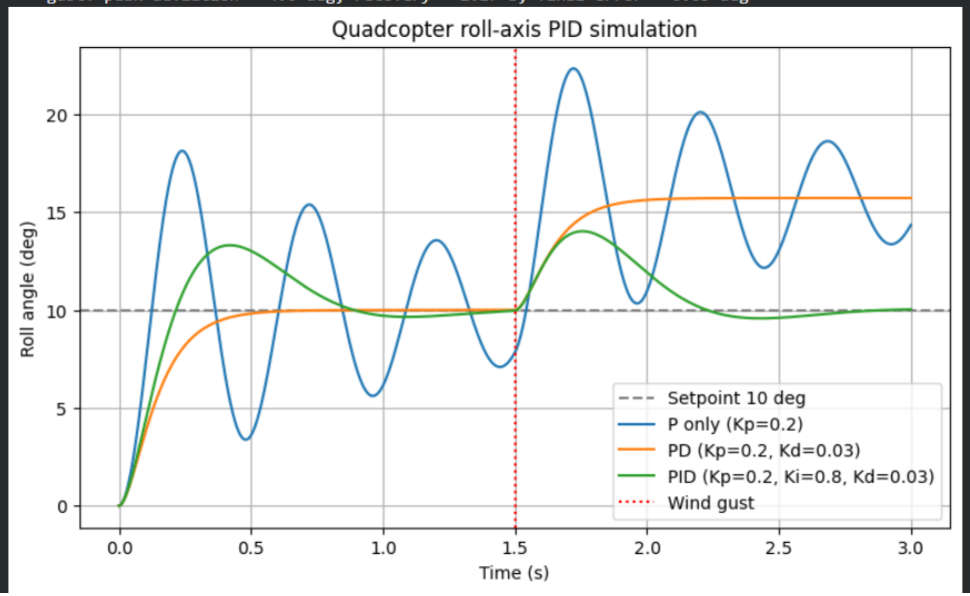
2.0 mm trial: displacement, max 2.491 mm

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PID Simulation: Roll Axis

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```
P only (Kp=0.2): overshoot = 81.4 %, settling time = 1.50 s
    gust: peak deviation = 12.4 deg, recovery = not recovered, final error = 4.35 deg
PD (Kp=0.2, Kd=0.03): overshoot = 0.0 %, settling time = 0.49 s
    gust: peak deviation = 5.7 deg, recovery = not recovered, final error = 5.73 deg
PID (Kp=0.2, Ki=0.8, Kd=0.03): overshoot = 33.1 %, settling time = 1.30 s
    gust: peak deviation = 4.0 deg, recovery = 1.17 s, final error = 0.03 deg
```



10 deg roll step with a 0.02 N.m wind gust at 1.5 s

Controller	Overshoot	Settling	Gust peak	Final error
P only (Kp 0.2)	81.4 %	1.50 s*	12.4 deg	4.35 deg
PD (Kd 0.03)	0.0 %	0.49 s	5.7 deg	5.73 deg
PID (Ki 0.8, Kd 0.03)	33.1 %	1.30 s	4.0 deg	0.03 deg

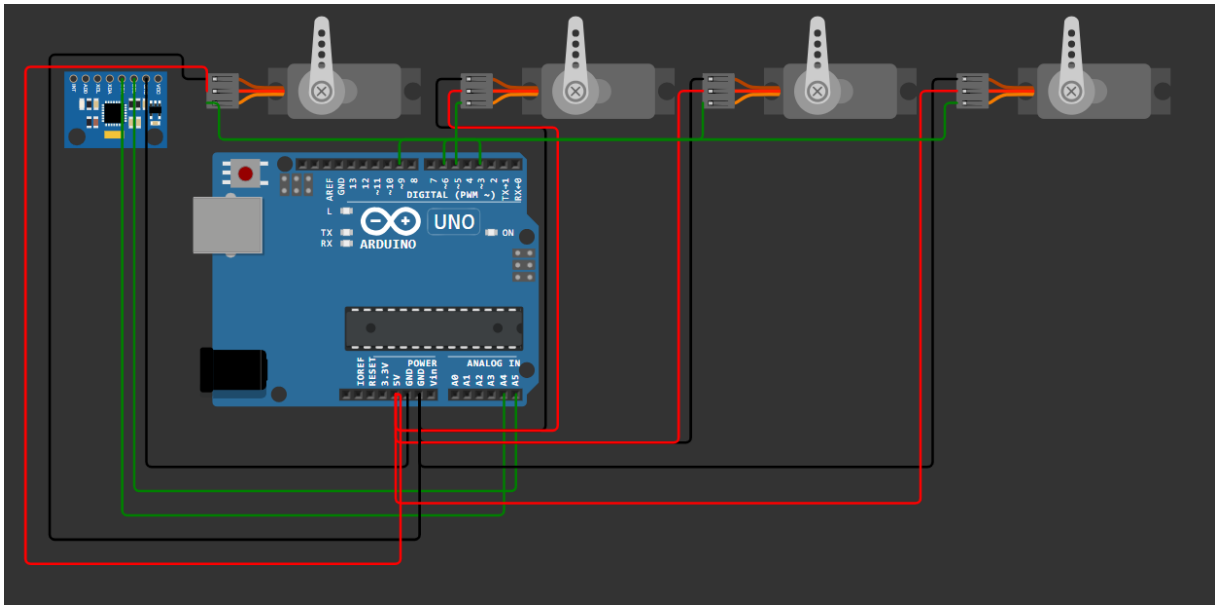
** Did not settle within the 1.5 s window (oscillates).*

- PD gives fast, non-overshooting step response but leaves a steady error under the gust.
- Adding the integral term removes the gust error (0.03 deg) at the cost of 33 % overshoot. Further tuning (lower Ki) was not done.
- Simplified single-axis model. Damping b = 0.002 and max torque 0.3 N.m are assumed. Inertia 1.17e-3 kg.m2 is a placeholder, not from the final CAD.

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Electronics Schematic and Assumptions

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Wokwi: Arduino Uno + MPU6050 (I2C: SDA A4, SCL A5) + 4 servos as ESC placeholders (PWM pins 3, 5, 6, 9)

Assumptions and limits

- Base geometry is reference geometry from GrabCAD.
- Component masses are estimates, not datasheet values.
- ABS yield strength assumed as 30 MPa (conservative, 3D-printed).
- Single static load case: 10 N per motor.
- PID model is single-axis and simplified, with a placeholder inertia.
- Servos stand in for ESCs (Wokwi has no ESC part). Real ESCs would be powered from the battery, not the Arduino 5 V pin.
- Hardware not built. All work is design and simulation only.